



Sustainable Smart Personal Assistant for Responsible Consumption

Presented by

**ITMCU Indonesia**

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Project Vision

# The Next Level of Eco-Conscious AI for Programming Education

# AI is Powerful But Costly

## The Vision

**Toward an Intelligent,  
Efficient, and Sustainable  
Educational Chatbot**

## Challenges in AI-Driven Education

- Rapid AI adoption in higher education
- AI integrated into students' workflows
- Reduced reflective reasoning
- High computational cost per inference
- Redundant AI-generated responses

## Opportunities

- Knowledge reuse improves system efficiency.
- Semantic retrieval reduces redundant AI inference.
- Gamification encourages more responsible and efficient AI usage.

# AI is Smart. But Are We Using It Responsibly?

## Current Research Focus

- Usability
- Ethics
- Learning outcomes

## What It Lacks

- Environmental awareness
- Sustainable usage behavior
- Practical system-level interventions

## Behavioral Insights

- Students have limited awareness of computational costs
- AI is often used for convenience rather than thoughtful engagement

**The problem is not the AI, it's how we use it.**

# Environmental Impact: Real & Measurable

## Inference Footprint [1]

- Up to **29 Wh per prompt** for large models.
- Equivalent to the annual electricity consumption of **35,000 households**.

## Training Impact [2]

- 493 metric tons** of CO<sub>2</sub> emissions.
- 2.7 million liters** of water consumed.

## Critical Gap:

There is no system that actively shapes responsible AI usage behavior.

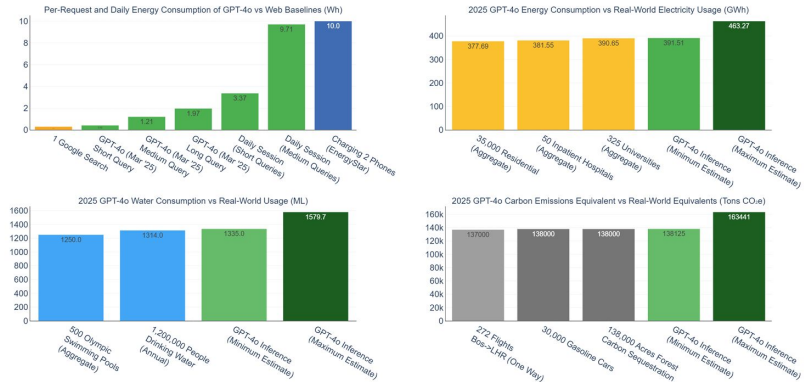


Figure 5: (Top Left) Per-query and daily energy consumption of GPT-4o. (Top Right) Estimated total annual energy usage of GPT-4o in 2025. (Bottom Left) The estimated 2025 annual water consumption of GPT-4o. (Bottom Right) The estimated 2025 annual carbon emissions of GPT-4o.

[1] N. Jegham et al., "How Hungry is AI? Benchmarking Energy, Water, and Carbon Footprint of LLM Inference," arXiv preprint arXiv:2505.09598, 2025.

[2] J. Morrison et al., "Holistically Evaluating the Environmental Impact of Creating Language Models," arXiv preprint arXiv:2503.05804, 2025.

# Introducing S-SPARC

More mindful. More efficient. More sustainable.

## Traditional AI (e.g., ChatGPT)

- Passive response system
- No control over user behavior
- No visibility of environmental impact
- Encourages convenience-driven usage

## S-SPARC

- Behavior-shaping AI system
- Real-time environmental tracking
- Gamified responsible usage
- Promotes mindful and efficient interaction

**Not just answering questions, shaping responsible AI usage**

# Intelligent Control & Sustainability

Three core pillars of the S-SPARC ecosystem

## Sustainability Tracker

- Monitors energy, carbon, and water usage
- Provides real-time environmental impact feedback
- Raises awareness of AI resource consumption

## Gamification

- Rewards efficient and mindful usage
- Leaderboard and feedback system
- Encourages responsible AI interaction

## Automated Evaluation

- Checks code quality and correctness
- Uses semantic similarity and static analysis
- Ensures reliable learning outcomes

**Not just smart, but responsible and sustainable**

# Technical Architecture

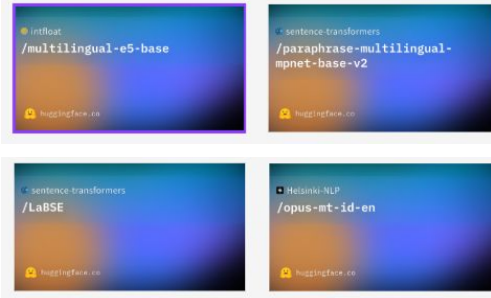
## Backend Stack



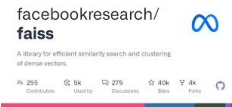
## Frontend Stack



## Pre-trained Models



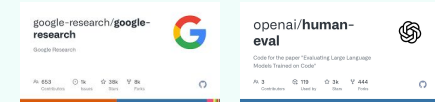
## Vector Search



## Database



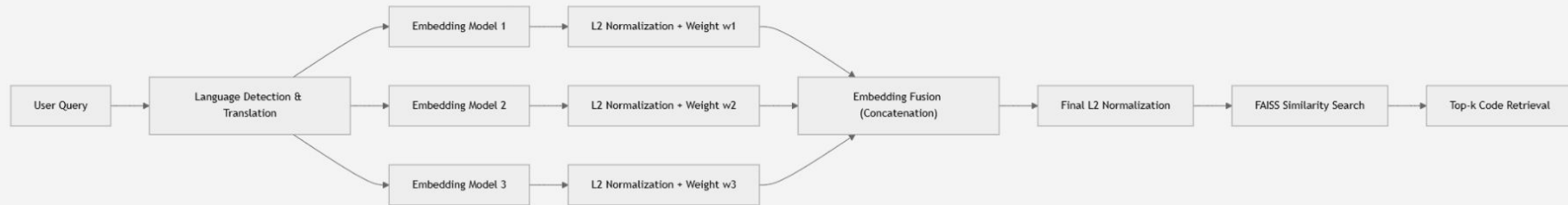
## Knowledge Base



## Emissions Tracker



# Semantic Retrieval Pipeline



## Language Processing

- Language detection
- English translation for better quality

## Weighted Multi-Model

- L2 normalization
- Weighted embeddings

## Embedding Fusion

- Weighted concatenation
- Complementary semantics

## Retrieval

- Final normalization
- FAISS similarity search

## Why Two Normalizations?

- **Before fusion:** Align embedding scales across different models
- **After fusion:** Ensure stable cosine similarity in FAISS



# Gamification Mechanisms

Gamification promotes efficient chatbot usage while supporting Green AI without compromising learning effectiveness.

## Objectives

- Encourage efficient and responsible AI usage
- Raise awareness of computational resource consumption
- Promote Green AI in education

## Scoring Mechanism

- Token usage recorded for every interaction
- Dynamic Threshold  $1.1 \times \text{Avg}(\text{Token})$ 
  - $\leq \text{Threshold}$   
Max score
  - $\geq \text{Threshold}$   
Lower score

## Anti Manipulation

- Relative scoring based on peer performance
- Prevent artificial token-saving strategies
- Scoring formula kept undisclosed

## Feedback & Rewards

- Leaderboard ranking for class visibility
- Rewards for top-performing students
- Increased motivation and engagement

# Automated Evaluation

Ensure the integrity and reliability of a self-growing knowledge base through a rigorous multi-metric evaluation framework.

## Evaluation Metrics

- **Alignment (40%)** – Prompt-to-code relevance
- **Logic (25%)** – Functional correctness
- **Semantic Similarity (20%)** – Benchmark match
- **Code Quality (10%)** – Readability
- **Static Analysis (5%)** – Syntax & complexity

## Composite Score

### F-Score

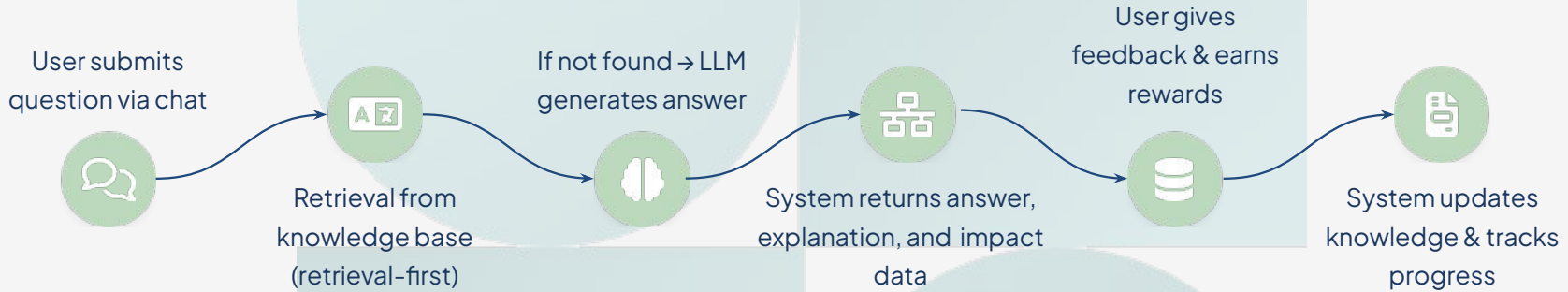
Weighted formulation:

$$F = 0.40(\text{Align}) + 0.25(\text{Logic}) + 0.20(\text{Semantic}) + 0.10(\text{Quality}) + 0.05(\text{Static})$$

## Design Rationale

- **65% Core Weight** on direct functionality (Alignment + Logic) to guarantee reliable execution outputs.
- **ISO/IEC 25010** empirical calibration standard used to align software quality attributes effectively.

# How It Works?



## Dynamic Threshold Gamification

Points are deducted proportionally for excessive usage

$$\text{Points}(u) = \begin{cases} 100, & u \leq T \\ \max(0, 100 + 100 \cdot \frac{T-u}{T}), & u > T \end{cases}$$

$T$  = dynamic threshold =  $1.10 \times u$   $u$  = weekly token usage of a learner

## Usage Policies

Minimum prompt lengths and a 1-minute rate limit to prevent "lazy prompting" and promote critical thinking.

## Environmental Impact Estimation

Measures environmental impact through token-based energy

$$E_{\text{kWh}} = \frac{t \cdot e(t)}{1000}, \quad \text{CO}_{2\text{e}_{\text{kg}}} = E_{\text{kWh}} \cdot \text{CIF}$$

$t$  = total tokens produced

$$\text{CIF} = 0.384 \text{ kgCO}_2 \text{ e} / \text{kWh}$$

$$e(t) = \begin{cases} 2.1775 \times 10^{-3} \text{ Wh/token}, & t \leq 400 \\ 1.5805 \times 10^{-3} \text{ Wh/token}, & 400 < t \leq 2000 \\ 4.2026 \times 10^{-4} \text{ Wh/token}, & t > 2000 \end{cases}$$



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# Experimental Design

## Approach

Quasi-experimental study

## Participants

- 55 undergraduate students
- Computer Science, Cohorts 2021–2024
- Machine Learning course

## Experimental Setting

- 7 laboratory sessions
- Varying task difficulty and complexity



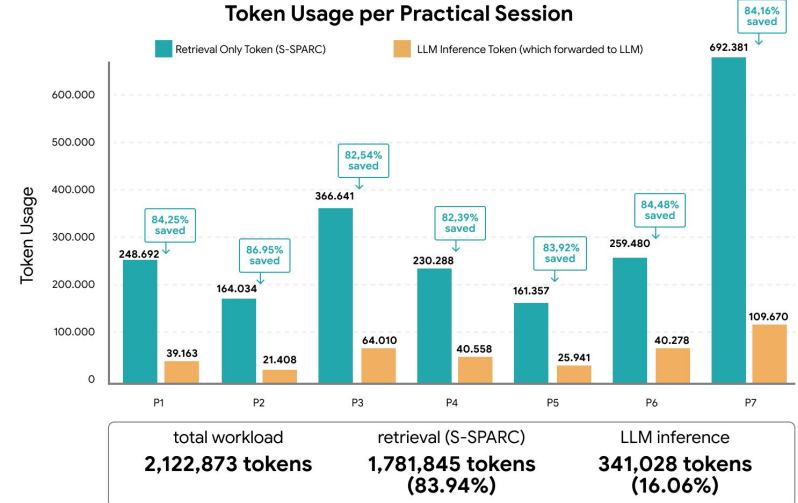
# System Efficiency

## 83.94% Efficiency

- Increased efficiency around **84%**
- Retrieval performance improves as the knowledge base grows
- Better recognition of recurring student queries

**6.2x Lower**  
Computational Load

**Smarter Retrieval,  
Lower Environmental Impact.**



### AI Has a Real Environmental Cost

#### Per API Call

0.0005 kWh energy  
0.015 kg CO<sub>2</sub>  
0.19 L water

#### Estimated Impact

26.29 kg CO<sub>2</sub> / task  
≈ 130–140 km driving  
300–500 kg CO<sub>2</sub> / semester

# Stability of the S-SPARC Reduction Rate

## Hypotheses & Setup

### Timeline Definition:

**Early:** Average reduction rate during first 3 weeks

**Late:** Average reduction rate during last 3 weeks

### Formulated Hypotheses:

**H<sub>0</sub>:** There is no significant difference in the reduction rate between the early and late phases of system usage.

**H<sub>1</sub>:** There is a significant difference in the reduction rate between the early and late phases of system usage.

| Parameter    | Early   | Late    |
|--------------|---------|---------|
| Mean         | 85.8%   | 86.9%   |
| Variance     | 0.00432 | 0.00413 |
| Observations | 55      | 55      |

$$t = -1.105 \quad p = 0.273 (> 0.05)$$

## Conclusion: Fail to reject H<sub>0</sub>

- No significant difference between early and late phases.
- S-SPARC maintained a consistently high reduction rate.
- Most inference requests were handled locally, minimizing direct GPT usage.

## Gamification Benefits

- Encourages more mindful and intentional AI usage among students
- Reduces non-essential and redundant queries, making interactions efficient
- Enhances self-regulation as students monitor their usage patterns

## Observed Behavioral Shift

- **From:** High-frequency, convenience-driven query patterns
- **Toward:** Selective, efficiency-oriented decision-making

Passive Users → **Responsible & Reflective AI Users**

# Bridging the Awareness Gap

## Key Insight

- An awareness gap exists between users and the true cost of AI
- Users often overlook computational and environmental impact
- Current systems do not actively guide responsible usage

## Our Contribution

- **S-SPARC** introduces system-level intervention
- Embeds responsibility directly into user interaction
- Transforms AI usage into a more mindful and efficient process

## Future Directions

- LMS integration for broader adoption
- Adaptive learning for personalized experience
- Expansion to multi-domain



# Thank You

*"The problem is not the AI, it's how we use it."*

**S-SPARC changes that.**

We are an AI innovation team from Maranatha Christian University, passionate about transforming education through artificial intelligence. Bringing together students and faculty with expertise in AI, software engineering, and education, we design and develop ethical, sustainable, and high-impact learning solutions for the future.

## Our main research themes:

- Academic integrity
- Educational computing
- Artificial intelligence

## Contact & Collaboration

Contact us for inquiries or collaboration!

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